Mathematical and Computational Methods

Exercise Sheet 2: Complex numbers I: algebra and geometry

This sheet accompanies *Lecture 2 (Complex numbers I: algebra and geometry)*. Every skill of the unit is exercised: Cartesian arithmetic, the conjugate and modulus, quadratics with complex roots, the Argand diagram, and the polar form. Work in radians throughout.

How to use this sheet. Problems are graded by difficulty—$\star$ drill (routine fluency), $\star\star$ core (standard, tutorial level) and $\star\star\star$ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ▶) and **further practice** for independent home study.

Solutions. This is the *student* sheet (problems only); a full worked solution sits after each problem but is hidden. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showsolutions{}\input{MCM_Exercises_Unit2}"`) and recompile.

Reminders. For $z = a + b\,\mathrm{i}$: conjugate $\bar{z} = a - b\,\mathrm{i}$, modulus $|z| = \sqrt{a^2 + b^2}$ with $|z|^2 = z\,\bar{z} = a^2 + b^2$, and division by $w \neq 0$ via $z/w = z\,\overline{w}/\,|w|^2$. Polar form $z = r(\cos\theta + \mathrm{i}\sin\theta)$ with $r = |z|$ and $\theta = \arg z$ (principal value $\operatorname{Arg} z \in (-\pi, \pi]$, fixed by the quadrant); products multiply moduli and add arguments.

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1 Warm-up drills

Exercise 1.1 *($\star$ drill)*. Simplify the powers of the imaginary unit

$$\mathrm{i}^3, \qquad \mathrm{i}^4, \qquad \mathrm{i}^{15}, \qquad \mathrm{i}^{2026}, \qquad \mathrm{i}^{-1}, \qquad \mathrm{i}^{-7}.$$

Exercise 1.2 *($\star$ drill)*. With $z = 3 + 2\mathrm{i}$ and $w = 1 - 5\mathrm{i}$, find $z + w$, $z - w$, zw and z^2.

Exercise 1.3 *($\star$ drill)*. For $z = 12 - 5\mathrm{i}$ write down $\bar{z}$ and compute $|z|$ and $z\,\bar{z}$.

Exercise 1.4 *($\star$ drill)*. For $z = -4 + 7\mathrm{i}$ state $\operatorname{Re} z$, $\operatorname{Im} z$, $\operatorname{Re}(\bar{z})$ and $\operatorname{Im}(\mathrm{i}z)$.

Exercise 1.5 *($\star$ drill)*. Write $\dfrac{4 + 3\mathrm{i}}{1 + 2\mathrm{i}}$ in Cartesian form $a + b\,\mathrm{i}$.

Exercise 1.6 *($\star$ drill)*. Solve $x^2 + 1 = 0$ and $x^2 + 25 = 0$ over $\mathbb{C}$.

Exercise 1.7 *($\star$ drill)*. Convert to polar form $r(\cos\theta + \mathrm{i}\sin\theta)$ with principal argument: $1 + \mathrm{i}$, $\quad 2\mathrm{i}$, $\quad -3$.

Exercise 1.8 *($\star$ drill)*. Convert $z = 2\big(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\big)$ to Cartesian form.

Exercise 1.9 *($\star$ drill)*. Let $z = 3\big(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\big)$ and $w = 2\big(\cos\frac{\pi}{12} + \mathrm{i}\sin\frac{\pi}{12}\big)$. Using the polar rules, find zw and z/w in Cartesian form.

Exercise 1.10 *($\star$ drill)*. On a single Argand diagram, plot the points $z_1 = 3 + \mathrm{i}$, $z_2 = -2 + 2\mathrm{i}$ and $z_3 = -1 - 3\mathrm{i}$, together with the sum $z_1 + z_2$. Label each point.

2 Tutorial problems

Problems marked ▶ are the core set for the 2-hour tutorial; the rest are for completion at home.

Exercise 2.1 *(▶ ⋆⋆ core)***.** Let $z = 2 + 3\mathrm{i}$ and $w = 1 - 4\mathrm{i}$ (as in the notes).

(a) Compute zw and z/w in Cartesian form.

(b) Verify the conjugate rule $\overline{zw} = \overline{z}\,\overline{w}$.

(c) Verify the modulus rule $|zw| = |z|\,|w|$.

Exercise 2.2 *(⋆⋆ core)***.** *(Maths stream — a general proof.)* Let $z = a + b\,\mathrm{i}$ and $w = c + d\,\mathrm{i}$ with $a, b, c, d \in \mathbb{R}$. Prove the conjugate-of-a-product rule $\overline{zw} = \overline{z}\,\overline{w}$ directly from the definition $\overline{a + b\mathrm{i}} = a - b\mathrm{i}$.

Exercise 2.3 *(▶ ⋆⋆ core)***.** Solve over $\mathbb{C}$, and confirm that the roots form a conjugate pair:

(a) $x^2 + 2x + 10 = 0$; (b) $x^2 - 4x + 13 = 0$; (c) $2x^2 + x + 1 = 0$.

Exercise 2.4 *(⋆⋆ core)***.** Find the real numbers x, y satisfying $(x + y\,\mathrm{i})(3 - \mathrm{i}) = 7 + \mathrm{i}$.

Exercise 2.5 *(▶ ⋆⋆ core)***.** Let $z = 3 + 2\mathrm{i}$ and $w = -1 + 2\mathrm{i}$. Plot z, w, $\overline{z}$, $z + w$, $z - w$ on an Argand diagram, and compute the distance $|z - w|$ between the points z and w.

Exercise 2.6 *(▶ ⋆⋆ core)***.** *(Physics — AC circuits.)* A series circuit has complex impedance $Z = 4 + 3\mathrm{i}$ (in ohms). It is driven by a voltage $V = 10$ (volts, taken as a real reference).

(a) Find $|Z|$ and the phase angle $\operatorname{Arg} Z$.

(b) Compute the current $I = V/Z$ in Cartesian form and its magnitude $|I|$.

Exercise 2.7 *(⋆⋆ core)***.** *(Data science — distance between records.)* Two data points in the plane are encoded as complex numbers $p = 2 + 5\mathrm{i}$ and $q = 6 + 2\mathrm{i}$. Compute the Euclidean distance $|p - q|$ between them and the midpoint $\frac{1}{2}(p + q)$.

Exercise 2.8 *(▶ ⋆⋆ core)***.** Convert each number to polar form, giving the *principal* argument $\operatorname{Arg} z \in (-\pi, \pi]$. Sketch each to fix the quadrant.

$$\text{(a)}\ -1 + \mathrm{i}; \qquad \text{(b)}\ -2 - 2\sqrt{3}\,\mathrm{i}; \qquad \text{(c)}\ \sqrt{3} - \mathrm{i}; \qquad \text{(d)}\ -5.$$

Exercise 2.9 *(⋆⋆ core)***.** Let $z = \sqrt{2}\big(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\big)$ and $w = 2\big(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\big)$. Find zw and z/w, giving each in both polar and Cartesian form.

Exercise 2.10 *(▶ ⋆⋆ core)***.** (a) For $z = 2 + \mathrm{i}$, compute $\mathrm{i}z$ and confirm geometrically that multiplying by i is a quarter-turn anticlockwise that preserves length.

(b) Which single complex number rotates a point by $\frac{\pi}{2}$ anticlockwise *and* doubles its distance from the origin?

(c) Apply a clockwise quarter-turn to $3 + \mathrm{i}$ (i.e. multiply by the number of modulus 1 and argument $-\frac{\pi}{2}$).

Exercise 2.11 *(⋆⋆ core)***.** Compute $(1 + \mathrm{i})^2$ and $(1 + \mathrm{i})^4$ by direct multiplication, and hence find $\big|(1 + \mathrm{i})^{10}\big|$ without expanding.

Exercise 2.12 *(▶ ⋆⋆ core)***.** *(Conceptual — true or false, with a one-line reason.)*

(a) For every $z \in \mathbb{C}$, $z^2 \geq 0$.

(b) $|z| = |\bar{z}|$ for all z.

(c) If $z + \bar{z} = 0$ then z is purely imaginary or zero.

(d) $\operatorname{Re}(zw) = \operatorname{Re}(z)\operatorname{Re}(w)$ for all z, w.

(e) There is an ordering $<$ on $\mathbb{C}$ compatible with its arithmetic — that is, one respecting $+$ and $\times$ (if $x > 0$ and $y > 0$ then $x + y > 0$ and $xy > 0$), so that every nonzero square is positive.

Exercise 2.13 *($\star\star\star$ challenge)*. *(Spot the error.)* A student writes the following two lines. Find and correct each mistake.

$$\text{(i)}\ \frac{2+3\mathrm{i}}{1+\mathrm{i}} = \frac{2}{1} + \frac{3}{1}\mathrm{i} = 2 + 3\mathrm{i}; \qquad \text{(ii)}\ \sqrt{-4}\,\sqrt{-9} = \sqrt{(-4)(-9)} = \sqrt{36} = 6.$$

3 Further practice (home)

Exercise 3.1 *($\star\star$ core)*. Evaluate the geometric-looking sum $\sum_{k=0}^{2026} \mathrm{i}^k = 1 + \mathrm{i} + \mathrm{i}^2 + \cdots + \mathrm{i}^{2026}$.

Exercise 3.2 *($\star$ drill)*. Compute in Cartesian form: (a) $(3-2\mathrm{i}) + (-1+5\mathrm{i})$; (b) $(2+\mathrm{i})(2-\mathrm{i})$; (c) $(1-\mathrm{i})^2$; (d) $\mathrm{i}(3-\mathrm{i})$.

Exercise 3.3 *($\star$ drill)*. For $z = -3 + 4\mathrm{i}$ find $\bar{z}$, $|z|$, $\frac{1}{z}$ and $\frac{z}{\bar{z}}$.

Exercise 3.4 *($\star\star$ core)*. Simplify to Cartesian form: (a) $\frac{1+2\mathrm{i}}{3-4\mathrm{i}}$; (b) $\frac{\mathrm{i}}{1+\mathrm{i}}$.

Exercise 3.5 *($\star\star$ core)*. Solve over $\mathbb{C}$: (a) $x^2 + 9 = 0$; (b) $x^2 - 6x + 25 = 0$; (c) $4x^2 + 9 = 0$; (d) $x^2 + x + 1 = 0$.

Exercise 3.6 *($\star\star$ core)*. Given that $1 + 2\mathrm{i}$ is a root of $x^3 - 4x^2 + 9x - 10 = 0$ (real coefficients), find the other two roots.

Exercise 3.7 *($\star\star$ core)*. Without converting to Cartesian form, evaluate $\left|\frac{(3+4\mathrm{i})(5-12\mathrm{i})}{1+\mathrm{i}}\right|$.

Exercise 3.8 *($\star\star$ core)*. Describe geometrically (and sketch) the set of points $z \in \mathbb{C}$ satisfying

$$\text{(a)}\ |z| = 2; \qquad \text{(b)}\ |z-1| = 3; \qquad \text{(c)}\ |z-\mathrm{i}| = |z+\mathrm{i}|\,.$$

Exercise 3.9 *($\star\star$ core)*. Sketch each *region* in the Argand plane (shade it, and show clearly whether the boundary is included):

$$\text{(a)}\ \{\, z : |z-1| < 2 \,\}; \qquad \text{(b)}\ \{\, z : 1 \le |z| \le 2 \,\}; \qquad \text{(c)}\ \{\, z : \operatorname{Re} z \ge 0,\ \operatorname{Im} z \ge 0 \,\}.$$

Exercise 3.10 *($\star\star$ core)*. Convert to Cartesian form:

$$\text{(a)}\ 6\big(\cos\tfrac{\pi}{6} + \mathrm{i}\sin\tfrac{\pi}{6}\big); \qquad \text{(b)}\ 2\big(\cos\tfrac{2\pi}{3} + \mathrm{i}\sin\tfrac{2\pi}{3}\big);$$
$$\text{(c)}\ 5(\cos\pi + \mathrm{i}\sin\pi); \qquad \text{(d)}\ \sqrt{2}\big(\cos(-\tfrac{\pi}{4}) + \mathrm{i}\sin(-\tfrac{\pi}{4})\big).$$

Exercise 3.11 *($\star\star$ core)*. Let $z = 4\big(\cos\frac{5\pi}{6} + \mathrm{i}\sin\frac{5\pi}{6}\big)$ and $w = 2\big(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\big)$. Find zw and z/w in Cartesian form.

Exercise 3.12 *($\star\star$ core)*. Compute the product $(1+\mathrm{i})(\sqrt{3}+\mathrm{i})$ in two ways — by Cartesian expansion and by the polar rule — and hence write down its modulus and argument.

Exercise 3.13 *(⋆⋆ core)*. *(Data science.)* Encode two real 2-vectors as $u = 3 + 4\mathrm{i}$ and $v = 4 + 3\mathrm{i}$. Show that the dot product of $(3, 4)$ and $(4, 3)$ equals $\operatorname{Re}(u\,\overline{v})$, and find $|u\,\overline{v}|$.

Exercise 3.14 *(⋆⋆ core)*. *(Physics — phasors / rotating vectors.)* Two voltage phasors of the same frequency are $V_1 = 6 + 8\mathrm{i}$ and a second phasor V_2 of equal amplitude but rotated a quarter-turn ($90°$) anticlockwise, i.e. $V_2 = \mathrm{i}V_1$. Find V_2 and the resultant $V = V_1 + V_2$ in Cartesian form, the resultant amplitude $|V|$, and show that $|V| = \sqrt{2}\,|V_1|$.

Exercise 3.15 *(⋆⋆⋆ challenge)*. Prove the *parallelogram law*: for all $z, w \in \mathbb{C}$,

$$|z + w|^2 + |z - w|^2 = 2\,|z|^2 + 2\,|w|^2 .$$

Exercise 3.16 *(⋆⋆⋆ challenge)*. Find all $z = x + y\,\mathrm{i}$ (with $x, y \in \mathbb{R}$) such that $z^2 = 3 + 4\mathrm{i}$, working directly in Cartesian form.

Exercise 3.17 *(⋆⋆ core)*. Find all complex z satisfying $z + 2\overline{z} = 6 - 3\mathrm{i}$.

Exercise 3.18 *(⋆⋆⋆ challenge)*. *(Rotation/geometry.)* Two vertices of an equilateral triangle lie at 0 and 2 on the real axis. Use multiplication by a complex number of modulus 1 to find the two possible positions of the third vertex.

Exercise 3.19 *(⋆⋆ core)*. *(Spot the error.)* A student claims “$|3 + 4\mathrm{i}| = 3 + 4 = 7$” and “$\operatorname{Arg}(-1 - \mathrm{i}) = \arctan(1) = \frac{\pi}{4}$”. Correct both.

Exercise 3.20 *(⋆⋆ core)*. Suppose $|z| = 2$, $\operatorname{Arg} z = \frac{\pi}{3}$, $|w| = 3$, $\operatorname{Arg} w = \frac{\pi}{4}$. Find the modulus and argument of zw, z/w, z^2 and $\overline{z}$; convert z and z^2 to Cartesian form.